# Model Card: Music Recommender Simulation

## 1. Model Name

Give your model a short, descriptive name.
MusicRecommender

## Collaborative Filtering vs. Content-Based Filtering

Collaborative filtering makes recommendations by analyzing patterns across many users. If thousands of people who liked Song A also liked Song B, the system will suggest Song B to a new listener who enjoyed Song A, even without knowing anything about the songs themselves. Content-based filtering ignores other users entirely and focuses on the attributes of the songs themselves. It compares features like genre, tempo, mood, and energy between songs a user already enjoys and new candidates in the catalog, recommending tracks that are musically similar to their established taste.

## Main Data Types Involved

These systems rely on two broad categories of data. Behavioral data captures how users interact with music, including things like likes, skips, replays, playlist additions, and listening duration, all of which signal preference without the user explicitly stating it. Attribute data describes the songs themselves, covering features such as tempo (in BPM), energy level, mood, genre, key, and valence (musical positivity). Together, these two data types let a platform build a picture of both what a user does and what a song sounds like, which is the foundation of any recommendation engine.

## Identify Key features

The songs.csv file contains ten tracks with ten attributes each: a numeric ID, title, artist, genre, mood, energy (on a 0.0 to 1.0 scale), tempo in BPM, valence, danceability, and acousticness. The most effective features for a content-based recommender would be genre and mood as categorical

matching points, since they directly capture the broad "feel" a user is after, while energy, valence, and danceability serve as strong numerical descriptors of a song's intensity and emotional tone. Tempo in BPM could also be useful for users who are sensitive to pace, such as someone building a workout playlist, and acousticness helps separate electronic or produced tracks from organic, instrument-driven ones. These features together give the system enough dimensions to distinguish between something like a chill lofi study track and a high-energy pop anthem, which aligns well with how people actually think about musical vibe.

**Mapping the Logic**

Genre match: +2.0

Mood match: +1.0

Energy proximity: up to +1.5

Valence proximity: up to +1.0

Danceability proximity: up to +1.0

To score a numerical feature like energy, you subtract the absolute difference between the song's value and the user's target from 1.0, then multiply by a weight. This means a song with an energy of 0.8 and a user target of 0.8 earns the full weight as a score, while a song with an energy of 0.4 earns much less, since the gap of 0.4 cuts deeply into the result. This approach rewards closeness rather than just high or low values, so a user who prefers moderate energy around 0.5 will not always get the most intense tracks pushed to the top.

The scoring rule and the ranking rule solve two different problems. The scoring rule is applied to one song at a time, taking the user's preferences and a single song's attributes and returning a number that represents how good a fit that song is. The ranking rule then applies that scoring rule across every song in the catalog, collects all the resulting numbers, and sorts them from highest to lowest so the system can identify the top

recommendations. Without the scoring rule there is no way to judge individual songs, and without the ranking rule there is no way to compare all those judgments against each other and decide which songs are actually worth suggesting.

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## 2. Intended Use

Describe what your recommender is designed to do and who it is for.

Prompts:

- What kind of recommendations does it generate
- What assumptions does it make about the user
- Is this for real users or classroom exploration

This recommender is designed to suggest songs from a small catalog that best match a user's stated taste preferences, such as their favorite genre, mood, and energy level. It assumes the user can describe their preferences in advance rather than learning from their listening behavior over time. It is built for classroom exploration and is not intended for production use on a real streaming platform.

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## 3. How the Model Works

Explain your scoring approach in simple language.

Prompts:

- What features of each song are used (genre, energy, mood, etc.)
- What user preferences are considered
- How does the model turn those into a score
- What changes did you make from the starter logic

Avoid code here. Pretend you are explaining the idea to a friend who does not program.

The system looks at five things for every song: its genre, its mood, its energy level, its valence (how positive or bright it sounds), and its danceability. The user provides a favorite genre, a favorite mood, and target numbers for energy, valence, and danceability. For genre and mood the system either awards points or it does not, since those are yes or no matches. For the numerical features it measures how close the song is to the user's target and awards more points the closer the match, so a user who wants moderate energy will not always get the most intense songs pushed to the top. All the points are added together and the songs with the highest totals are recommended first.

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**4. Data**

Describe the dataset the model uses.

Prompts:

- How many songs are in the catalog
- What genres or moods are represented
- Did you add or remove data
- Are there parts of musical taste missing in the dataset

The catalog contains 20 songs total, 10 from the starter file and 10 that were generated to add variety. The genres represented include pop, lofi, rock, ambient, jazz, synthwave, indie pop, country, edm, kpop, folk, blues, metal, bossa nova, chiptune, rnb, and ambient. Moods range from happy and chill to intense, moody, melancholy, aggressive, and romantic. The dataset is still very small for a real recommender and is missing entire categories of musical taste like classical, hip hop, reggae, and Latin music, which means users with those preferences will never get a strong genre match.

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**5. Strengths**

Where does your system seem to work well

Prompts:

- User types for which it gives reasonable results
- Any patterns you think your scoring captures correctly
- Cases where the recommendations matched your intuition

The system works best for users whose tastes align with the most common genres in the catalog, particularly pop and lofi listeners. The proximity scoring for energy and valence does a good job of separating high-intensity tracks from calm ones, so a chill lofi profile consistently surfaces quiet low-energy songs while a high-energy edm profile pulls the most intense tracks to the top. The reasons output also makes it easy to understand exactly why each song ranked where it did, which is something real recommenders rarely show the user.

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**6. Limitations and Bias**

Where the system struggles or behaves unfairly.

Prompts:

- Features it does not consider
- Genres or moods that are underrepresented
- Cases where the system overfits to one preference
- Ways the scoring might unintentionally favor some users

The biggest limitation is that genre match carries so much weight that two songs with very similar energy and mood profiles can rank far apart simply because one matches the user's favorite genre and the other does not. The

dataset is also small enough that some profiles will see the same songs appear repeatedly across different users, since there may only be one or two songs that match a niche genre. The system also ignores tempo entirely in the scoring, so a user who wants fast music for running could end up with slow songs that happen to match their genre and mood. Additionally the catalog has no hip hop or Latin songs at all, meaning any user with those preferences will receive recommendations driven entirely by numerical proximity rather than any categorical match.

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## 7. Evaluation

How you checked whether the recommender behaved as expected.

Prompts:

- Which user profiles you tested
- What you looked for in the recommendations
- What surprised you
- Any simple tests or comparisons you ran

No need for numeric metrics unless you created some.

Three user profiles were tested: a Happy Pop Fan targeting high energy and positive valence, a Chill Lofi Listener targeting low energy and moderate valence, and a High-Energy EDM Fan targeting maximum energy and danceability. The pop profile consistently surfaced Sunrise City and Rooftop Lights at the top, which matched intuition well. The lofi profile correctly pulled Library Rain and Midnight Coding to the front. The most surprising result was that the EDM profile sometimes ranked non-EDM songs highly when their energy and danceability scores were close enough to compensate for the missing genre match, which revealed that the numerical weights are strong enough to partially override the categorical ones.

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## 8. Future Work

Ideas for how you would improve the model next.

Prompts:

- Additional features or preferences
- Better ways to explain recommendations
- Improving diversity among the top results
- Handling more complex user tastes

The most useful improvement would be adding a tempo preference to the user profile and including it in the scoring, since BPM is already in the dataset but currently unused. A second improvement would be adding a diversity rule that prevents the same artist or genre from appearing more than twice in the top five results, which would reduce the feeling of being stuck in a loop. Third, replacing the fixed weights with user-adjustable sliders would let people decide for themselves whether genre or energy matters more to them, making the system much more flexible without changing the underlying math.

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## 9. Personal Reflection

A few sentences about your experience.

Prompts:

- What you learned about recommender systems
- Something unexpected or interesting you discovered
- How this changed the way you think about music recommendation apps

The biggest learning moment was realizing how much a small dataset shapes the results in ways that have nothing to do with the algorithm itself. Even with a well-designed scoring function, a catalog with gaps in certain

genres will always produce biased recommendations for users who like those genres. It was also surprising how much the simple weighted scoring felt like a real recommendation when the weights were tuned well, even though the logic behind it is just addition and subtraction. This changed the way I think about apps like Spotify, because it made clear that a lot of what feels like intelligence in a recommender is actually just careful decisions about what features to measure and how much weight to give each one.

```
Loaded songs: 20

==================================================
Profile: Happy Pop Fan
==================================================
1. Sunrise City by Neon Echo
   Score:   6.42
   Reasons: genre match (+2.0), mood match (+1.0), energy proximity (+1.47), valence proximity (+0.96), danceabil
ity proximity (+0.99)
--------------------------------------------------
2. Gym Hero by Max Pulse
   Score:   5.19
   Reasons: genre match (+2.0), energy proximity (+1.3), valence proximity (+0.97), danceability proximity (+0.92
)
--------------------------------------------------
3. Rooftop Lights by Indigo Parade
   Score:   4.41
   Reasons: mood match (+1.0), energy proximity (+1.44), valence proximity (+0.99), danceability proximity (+0.98
)
--------------------------------------------------
4. Pixel Dreams by Chiptune Ghost
   Score:   3.28
   Reasons: energy proximity (+1.33), valence proximity (+0.96), danceability proximity (+0.99)
--------------------------------------------------
5. Slow Burn by Ember Trails
   Score:   3.15
   Reasons: energy proximity (+1.24), valence proximity (+0.94), danceability proximity (+0.97)
--------------------------------------------------
```

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==================================================
Profile: Chill Lofi Listener
==================================================
1. Library Rain by Paper Lanterns
   Score:   6.42
   Reasons: genre match (+2.0), mood match (+1.0), energy proximity (+1.5), valence proximity (+0.95), danceabili
ty proximity (+0.97)
--------------------------------------------------
2. Midnight Coding by LoRoom
   Score:   6.32
   Reasons: genre match (+2.0), mood match (+1.0), energy proximity (+1.4), valence proximity (+0.99), danceabili
ty proximity (+0.93)
--------------------------------------------------
3. Focus Flow by LoRoom
   Score:   5.33
   Reasons: genre match (+2.0), energy proximity (+1.42), valence proximity (+0.96), danceability proximity (+0.9
5)
--------------------------------------------------
4. Spacewalk Thoughts by Orbit Bloom
   Score:   4.16
   Reasons: mood match (+1.0), energy proximity (+1.4), valence proximity (+0.9), danceability proximity (+0.86)
--------------------------------------------------
5. Coffee Shop Stories by Slow Stereo
   Score:   3.30
   Reasons: energy proximity (+1.47), valence proximity (+0.84), danceability proximity (+0.99)
--------------------------------------------------
```

```
==================================================
Profile: High-Energy EDM Fan
==================================================
1. Neon Carnival by Pixel Riot
   Score:   6.47
   Reasons: genre match (+2.0), mood match (+1.0), energy proximity (+1.5), valence proximity (+0.98), danceabili
ty proximity (+0.99)
--------------------------------------------------
2. Gym Hero by Max Pulse
   Score:   3.27
   Reasons: energy proximity (+1.47), valence proximity (+0.87), danceability proximity (+0.93)
--------------------------------------------------
3. Sunrise City by Neon Echo
   Score:   3.08
   Reasons: energy proximity (+1.3), valence proximity (+0.94), danceability proximity (+0.84)
--------------------------------------------------
4. Rooftop Lights by Indigo Parade
   Score:   3.00
   Reasons: energy proximity (+1.22), valence proximity (+0.91), danceability proximity (+0.87)
--------------------------------------------------
5. Pixel Dreams by Chiptune Ghost
   Score:   2.83
   Reasons: energy proximity (+1.11), valence proximity (+0.86), danceability proximity (+0.86)
--------------------------------------------------
```